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ORIGINAL RESEARCH

FOCUS ON TRANSCATHETER AORTIC VALVE REPLACEMENT

Bleeding in Patients Undergoing Transfemoral Transcatheter Aortic Valve Replacement

Incidence, Trends, Clinical Outcomes, and Predictors

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ABSTRACT

BACKGROUND Bleeding is one of the most frequent complications in patients undergoing transcatheter aortic valve replacement (TAVR). Importantly, major bleeding is associated with poor clinical outcomes after TAVR. However, large studies on bleeding complications in the contemporary TAVR population are limited.

OBJECTIVES The aim of this study was to assess the incidence, temporal trends, clinical outcomes, and predictors of bleeding in patients undergoing transfemoral TAVR.

METHODS The CENTER2 study is a pooled patient-level database from 10 clinical studies including patients who underwent TAVR between 2007 and 2022.

RESULTS A total of 23,562 patients underwent transfemoral TAVR. The mean age was 81.5 ± 6.7 years, and 56% were women. Major bleeding within the first 30 days was observed in 1,545 patients (6.6%). Minor bleeding was reported in 1,143 patients (4.7%). Rates of major bleeding decreased from 11.5% in 2007-2010 to 5.5% in 2019-2022 ($P_{trend} < 0.001$). Dual antiplatelet therapy was associated with higher major bleeding rates compared with single antiplatelet therapy (12.2% vs 9.1%; OR: 1.40; 95% CI: 1.13-1.72; $P = 0.002$). Patients with major bleeding had increased mortality risk during the first 30 days (14.1% vs 4.3%; OR: 3.66; 95% CI: 3.11-4.31; $P < 0.001$) and during 1-year follow-up (27.8% vs 14.5%; HR: 1.50; 95% CI: 1.41-1.59; $P < 0.001$). Minor bleeding did not affect 1-year mortality risk (16.7% vs 14.5%; HR: 1.11; 95% CI: 0.93-1.32; $P = 0.27$). Predictors of major bleeding were female sex and peripheral vascular disease.

CONCLUSIONS Bleeding complications remain frequent and important in patients undergoing transfemoral TAVR. Increased mortality risk in major bleeding persists after the initial 30 days. (Cerebrovascular Events in Patients Undergoing Transcatheter Aortic Valve Implantation With Balloon-Expandable Valves Versus Self-Expandable Valves [CENTER]; NCT03588247) (J Am Coll Cardiol Intv 2023;16:2951-2962) © 2023 by the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

DAPT = dual antiplatelet therapy

OAC = oral anticoagulation

PCI = percutaneous coronary intervention

SAPT = single antiplatelet therapy

TAVR = transcatheter aortic valve replacement

VARC-2 = Valve Academic Research Consortium 2

Transcatheter aortic valve replacement (TAVR) has become the preferred treatment in older patients with severe aortic valve stenosis and is now exceeding the annual number of patients treated with surgical aortic valve replacement.[1,2] TAVR has been associated with lower rates of periprocedural major bleeding compared with conventional surgery.[3] However, bleeding is still among the most frequent complications of TAVR.[4] In general, patients with severe aortic valve stenosis undergoing TAVR harbor a high risk for bleeding because of pre-existing comorbidities. In addition, procedural characteristics play a major role in the occurrence of bleeding after TAVR.[5] Remarkably, large studies on temporal trends, clinical outcomes, and risk factors of bleeding in the contemporary TAVR population are limited. Therefore, the aim of this study was to assess incidence, temporal trends, clinical outcomes and predictors of bleeding complications in patients undergoing TAVR between 2007 and 2022.

METHODS

STUDY DESIGN. The CENTER2 study is an international collaboration on patients with aortic valve stenosis undergoing TAVR. The initial CENTER (Cerebrovascular Events in Patients Undergoing Transcatheter Aortic Valve Implantation With Balloon-Expandable Valves Versus Self-Expandable Valves) study included 12,381 patients undergoing transfemoral TAVR between 2007 and 2018. The design of the CENTER study has been described previously.[6] The patient-level database consists of 1 prospective clinical trial, 2 multicenter prospective registries, 3 national registries, and 4 single-center prospective registries, identified through a systematic search. In April 2022, collaborators were asked to add more contemporary patients to the database. An additional 13,390 patients who underwent TAVR between 2018 and 2022 were added to the database, creating the updated CENTER2 study. The study includes a worldwide population, as patients were treated in Spain, the United States, Italy, Israel, France, and Brazil. The collaborating centers provided a dedicated database including patient characteristics, echocardiographic measurements, procedural data, and long-term follow-up. Surgical risk profiles were estimated by Society of Thoracic Surgeons Predicted Risk of Mortality and divided into low risk (<4%), intermediate risk (4%-8%), and high risk (>8%). Indication for TAVR and selection of valve type and size were made by the multidisciplinary heart team at each participating center. Type and periprocedural management of antithrombotic treatment were determined in accordance with international guidelines. Oral anticoagulation (OAC) were stopped prior to the TAVR procedure, and the international normalized ratio was routinely checked before the procedure. Procedural anticoagulation was done with unfractionated heparin. However, 1.5% of patients were recruited from the interventional arm of the BRAVO-3 (Effect of Bivalirudin on Aortic Valve Intervention Outcomes 3) study and therefore procedurally treated with bivalirudin. All centers used suture-based closure devices. An overview of the included studies is provided in Supplemental Table S1. All individual studies were conducted according to the Declaration of Helsinki. Institutional Review Board approval was obtained at each center. All patients provided written informed consent at their local hospital. The study is registered at ClinicalTrials.gov (NCT03588247).

PATIENT POPULATION. The CENTER2 study consists of 25,771 patients undergoing TAVR, of whom 24,321 (94.4%) underwent transfemoral TAVR. Historically, nontransfemoral TAVR is associated with increased risk for bleeding due to procedural aspects (eg, surgical cutdown) and is usually performed in patients with increased risk profiles. Therefore, we included only patients treated with transfemoral TAVR in the present study. A total of 759 patients (3.1%) had

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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unknown 30-day bleeding status. The incidence of 30-day bleeding was available in 23,562 patients (96.9%) undergoing transfemoral TAVR.

STUDY DEFINITIONS AND OUTCOMES. All overt bleeding events were defined according to the Valve Academic Research Consortium-2 (VARC-2).[7] Overt bleeding was defined as any bleeding that was identified on physical examination or diagnostic testing, as recommended by VARC-2. In accordance with VARC-2, bleeding events were classified as life threatening, major, or minor. The primary outcome of the present study was major bleeding within 30 days after transfemoral TAVR, incorporating both life-threatening and major bleeding. The first bleeding occurring after TAVR was reported, and multiple events were not included in the analysis. Late bleeding events, defined as bleeding complications occurring after 30 days, were also reported. Bleeding events were subdivided into access-site and non-access-site bleeding. Access-site bleeding was defined as any bleeding related to either the location of device insertion or any other accessory access site used during the procedure. Other bleeding complications were considered non-access-site bleeding. Moreover, clinical outcomes in patients with major and minor bleeding complications were studied. Devices were considered as newer generation if they became commercially available after 2013. Supplemental Table S2 presents an overview of valve types used.

STATISTICAL ANALYSIS. Baseline continuous variables were visually tested for normality using distribution plots. Accordingly, variables are reported as mean ± SD or median (Q1-Q3), and differences were tested using Student's *t*-test or the Mann-Whitney *U* test. Baseline categorical variables are reported as frequencies and percentages, and differences were tested using the chi-square test. The incidence of major bleeding was assessed at 30 days. Changes in 30-day major and minor bleeding incidence over time were tested using the chi-square Mantel-Haenszel test for trend. Differences in bleeding rates for different subgroups were tested using the chi-square test, and ORs with 95% CIs were established using logistic regression. The reference category for ORs was patients without bleeding events.

Differences in clinical outcomes among patients with major, minor, or no bleeding were tested using the chi-square test, and ORs with 95% CIs were established using logistic regression. Time-to-mortality estimates were made using Cox regression, and HRs are reported. The reference category for HRs was patients without bleeding events. The effect of 30-day major bleeding on mortality after the initial acute postprocedural period was assessed using Cox regression analyses with landmark time at 30 and 90 days. Subgroup analysis was performed for access-site and non-access-site bleeding.

Baseline characteristics were explored as predictors of 30-day major bleeding using univariate logistic regression analysis. Predictors with *P* values <0.10 and predictors known from the literature were simultaneously entered in a multivariate logistic regression model. Multivariate predictors are reported as OR (95% CI).

All statistical tests were 2 tailed, and *P* values <0.05 were considered to indicate statistical significance. Calculations were performed using SPSS version 28.0 for Windows (SPSS).

RESULTS

BASELINE CHARACTERISTICS. A total of 23,562 patients undergoing transfemoral TAVR were included in the present study. The mean age was 81.5 ± 6.7 years, and 56% of patients were women. The median Society of Thoracic Surgeons Predicted Risk of Mortality score was 4.9% (Q1-Q3: 3.1%-8.5%). Balloon-expandable valves were used in 43% of patients, and 58% of patients were treated using newer generation valves. Median follow-up duration was 365 days (Q1-Q3: 40-653 days). **Table 1** presents baseline characteristics of the included cohort. Detailed information on antithrombotic treatment was available in 14,123 patients (58%). Antithrombotic treatment at baseline and periprocedural changes in antithrombotic treatment are displayed in **Figure 1** and Supplemental Table S3.

INCIDENCE OF BLEEDING. A total of 1,545 patients (6.6%) had major bleeding within the first 30 days after TAVR. Of these, 453 (29.4%) were classified as having life-threatening bleeding. An additional 1,143 patients (4.7%) experienced minor bleeding. Rates of major bleeding decreased from 11.5% in 2007-2010 to 5.5% in 2019-2022 ($P_{trend} < 0.001$). In contrast, the incidence of minor bleeding did not significantly decrease from 9.9% in 2007-2010 to 4.3% in 2019-2022 ($P_{trend} = 0.75$). Trends in major and minor bleeding incidence are displayed in **Figure 2**. The vast majority of major and minor bleeding complications occurred on the first day after TAVR, as presented in **Figure 3**. Late bleeding (>30 days after TAVR) was reported in an additional 72 patients (0.38%). Of these, 39 (54%) were major and 33 (46%) were minor. The median time between TAVR and late bleeding was 362 days (Q1-Q3: 96-366 days).

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TABLE 1 Baseline and Procedural Characteristics in Patients Undergoing Transfemoral Transcatheter Aortic Valve Replacement

	Total Population (N = 23,562)	Major Bleeding (n = 1,545)	Minor Bleeding (n = 1,143)	No Bleeding (n = 20,874)	*P* Value
Demographics					
Age, y	81.5 ± 6.7	81.9 ± 6.4	81.5 ± 6.2	81.4 ± 6.7	0.03
Women	13,168 (56)	999 (65)	635 (56)	11,534 (55)	<0.001
Body mass index, kg/m^2	27.5 ± 4.9	26.9 ± 4.7	27.4 ± 5.0	27.5 ± 4.9	<0.001
Medical history					
Stroke or TIA	2,361 (10)	144 (10)	126 (11)	2,091 (10)	0.40
Myocardial infarction	2,898 (13)	176 (12)	149 (14)	2,573 (13)	0.39
Previous PCI	4,702 (21)	322 (21)	250 (22)	4,130 (21)	0.31
Previous CABG	1,982 (9)	117 (8)	102 (9)	1,763 (9)	0.29
Hypertension	18,894 (81)	1,256 (82)	985 (87)	15,542 (81)	<0.001
Dyslipidemia	13,605 (59)	862 (57)	736 (65)	12,007 (59)	<0.001
Peripheral vascular disease	2,772 (12)	236 (16)	162 (14)	2,374 (12)	<0.001
Coronary artery disease	8,509 (38)	586 (39)	460 (41)	7,463 (37)	0.02
Atrial fibrillation	6,368 (28)	412 (27)	353 (31)	5,603 (28)	0.04
Renal failure	2,530 (12)	186 (13)	133 (13)	2,211 (12)	0.29
GFR, mL/min/1.73 m^2	52.6 (38.9-69.3)	51.2 (36.9-67.7)	52.6 (38.0-70.3)	52.8 (39.1-69.5)	0.03
NYHA functional class ≥III	8,544 (51)	495 (56)	365 (48)	7,286 (50)	0.02
Risk scores					
STS-PROM	4.9 (3.1-8.5)	5.2 (3.4-8.9)	5.2 (3.3-8.8)	4.8 (3.0-8.3)	<0.001
Logistic EuroSCORE	13.5 (8.5-21.4)	13.4 (8.6-21.7)	13.4 (8.4-21.4)	13.3 (8.4-21.3)	0.87
EuroSCORE II	3.6 (2.2-6.0)	4.0 (2.6-6.3)	3.7 (2.2-6.2)	3.6 (2.2-6.0)	0.002
Echocardiographic characteristics					
LVEF, %	56.9 ± 13.1	57.9 ± 12.7	58.5 ± 12.2	56.7 ± 13.2	<0.001
Peak gradient, mm Hg	77.8 ± 23.2	80.8 ± 23.0	77.3 ± 23.9	77.6 ± 23.2	<0.001
Mean gradient, mm Hg	49.0 ± 16.5	52.5 ± 17.4	48.1 ± 16.2	48.6 ± 16.3	<0.001
Aortic valve area, cm^2	0.67 ± 0.20	0.67 ± 0.20	0.69 ± 0.20	0.67 ± 0.20	0.006
Device type					
Balloon-expandable valve	10,069 (43)	704 (46)	432 (38)	8,933 (43)	<0.001
Newer generation valve	13,470 (58)	699 (47)	699 (62)	12,072 (58)	<0.001
Valve-in-valve	588 (3)	34 (3)	39 (4)	515 (3)	0.27
Valve size, mm	26 (25-29)	26 (23-19)	26 (25-29)	26 (25-29)	0.06
Sheath size					<0.001
≤14-F	6,813 (32)	346 (26)	389 (38)	6,078 (32)	
16-F	4,848 (23)	330 (25)	225 (22)	4,293 (23)	
18-F	8,405 (40)	615 (46)	367 (36)	7,423 (39)	
>18-F	1,228 (6)	55 (4)	47 (5)	1,126 (6)	

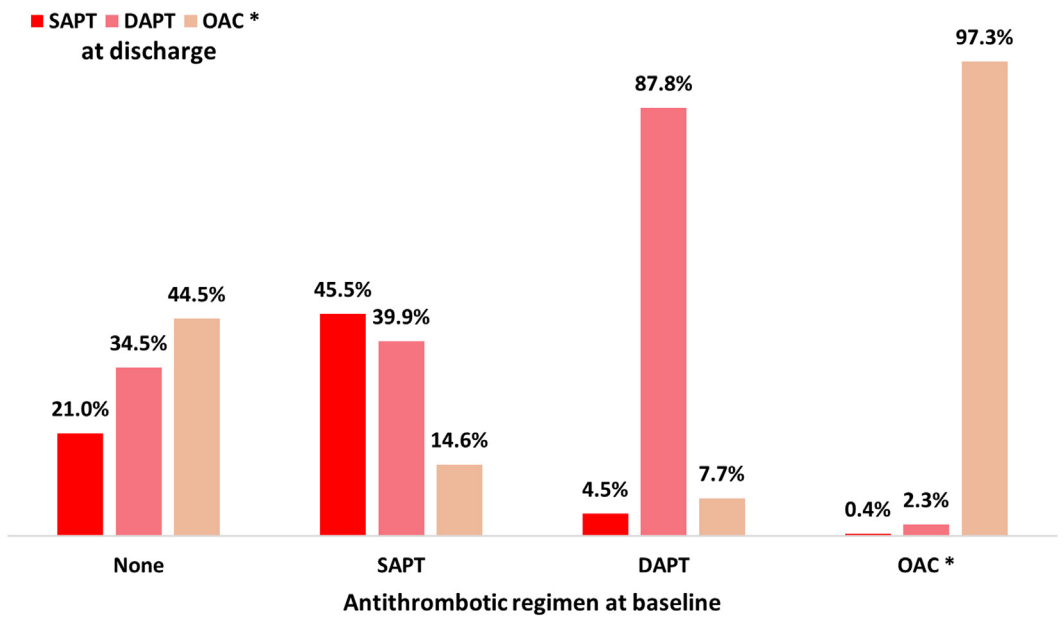
Values are mean ± SD, n (%), or median (Q1-Q3).

CABG = coronary artery bypass graft; EuroSCORE = European System for Cardiac Operative Risk Evaluation; GFR = glomerular filtration rate; LVEF = left ventricular ejection fraction; PCI = percutaneous coronary intervention; STS-PROM = Society of Thoracic Surgeon Predicted Risk of Mortality; TIA = transient ischemic attack.

The incidence of major bleeding within 30 days after TAVR was higher in women (8.0% vs 5.5%; OR: 1.48; 95% CI: 1.33-1.65; $P < 0.001$). Patients with intermediate and high surgical risk reported higher rates of major bleeding compared with those with low surgical risk (7.1% vs 6.8% vs 5.4%; $P = 0.002$). Major bleeding incidence was not different in patients with or without histories of atrial fibrillation (6.8% vs 7.1%; $P = 0.46$) or in patients with previous percutaneous coronary intervention (PCI) (7.2% vs 6.9%; $P = 0.51$). On the contrary, minor bleeding was more frequent in patients with atrial fibrillation (6.0% vs 5.2%; OR: 1.18; 95% CI: 1.03-1.34; $P = 0.02$). Patients with peripheral vascular disease had a higher incidence of major bleeding (9.0% vs 6.2%; OR: 1.39; 95% CI: 1.20-1.61; $P < 0.001$) and minor bleeding (6.5% vs 5.3%; OR: 1.24; 95% CI: 1.05-1.47; $P = 0.01$). Patients treated with balloon-expandable valves reported higher rates of major bleeding compared with patients treated with self-expandable valves (7.3% vs 6.6%; OR: 1.12; 95% CI: 1.01-1.24; $P = 0.03$) but lower rates of minor bleeding (4.3% vs 5.3%; OR: 0.81; 95% CI: 0.71-0.91; $P = 0.001$).

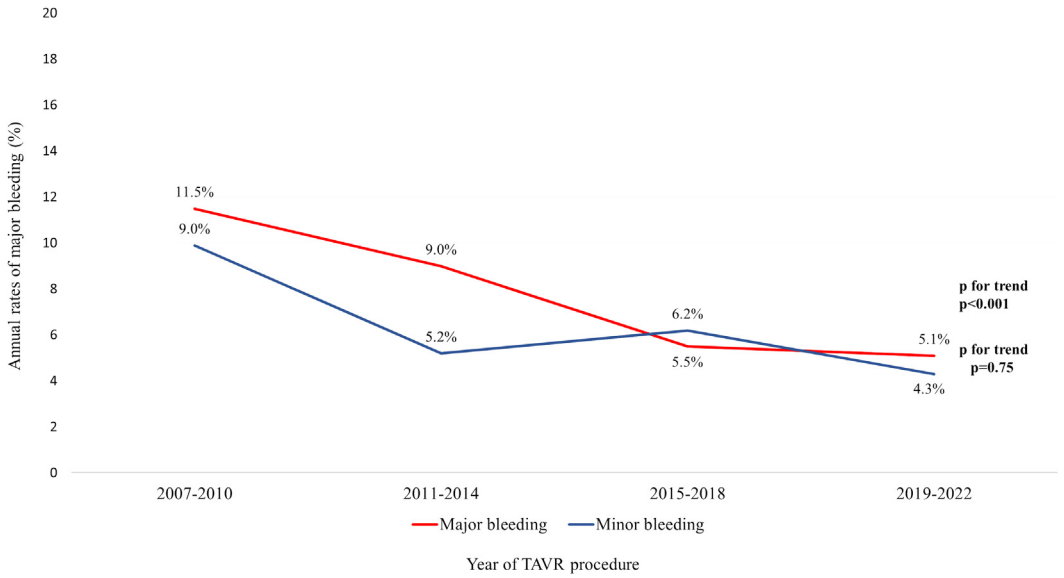
ANTITHROMBOTIC THERAPY. Dual antiplatelet therapy (DAPT) at baseline was associated with higher rates of 30-day major bleeding compared with single antiplatelet therapy (SAPT) (12.2% vs 9.1%; OR: 1.40; 95% CI: 1.13-1.72; $P = 0.002$). Patients on OAC (interrupted before the procedure) did not have significantly higher rates of major bleeding compared with those on SAPT (10.9% vs 9.1%; OR: 1.24; 95% CI:

FIGURE 1 Antithrombotic Treatment at Discharge Displayed for Different Baseline Antithrombotic Regimens



*OAC only or in combination with SAPT or DAPT. DAPT = dual antiplatelet therapy; OAC = oral anticoagulation; SAPT = single antiplatelet therapy.

FIGURE 2 Temporal Trends in Major and Minor Bleeding at 30 Days in Patients Undergoing Transfemoral TAVR



TAVR = transcatheter aortic valve replacement.

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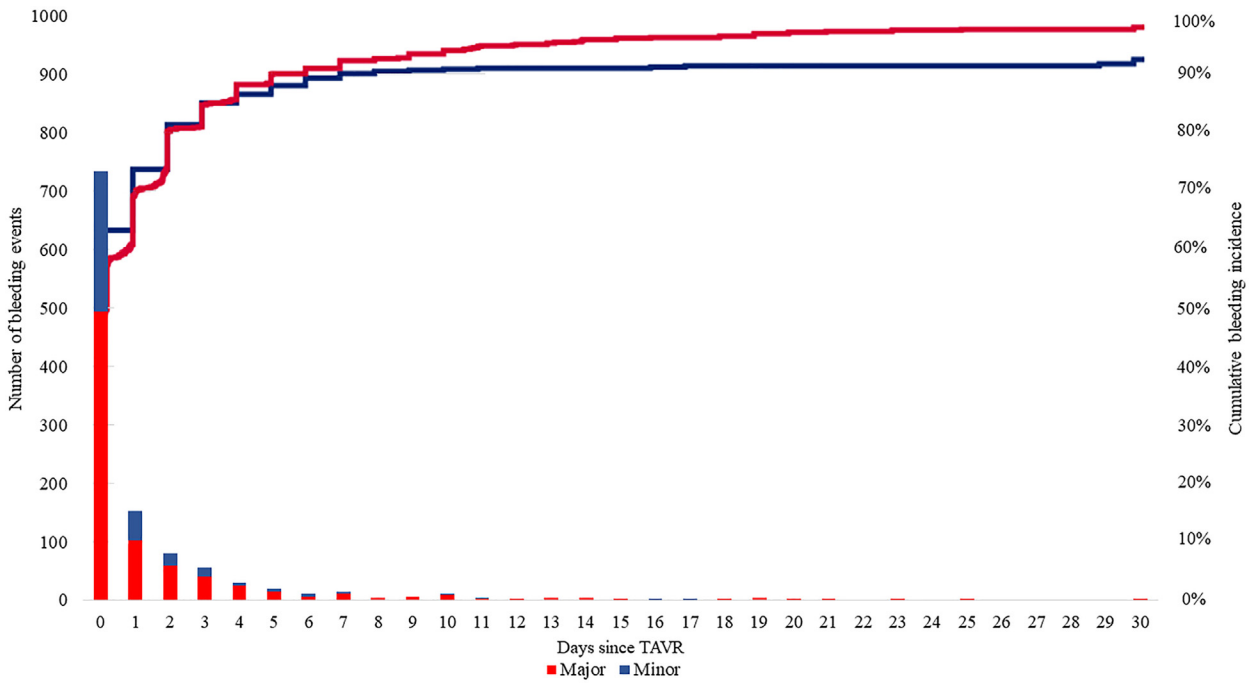
FIGURE 3 Number of Bleeding Events According to Days Since Transfemoral TAVR for Major and Minor Bleeding



TAVR = transcatheter aortic valve replacement.

0.94-1.63; $P = 0.13$) or on DAPT at 30 days (10.9% vs 12.2%; OR: 0.87; 95% CI: 0.66-1.15; $P = 0.33$).

Minor bleeding at 30 days was higher in patients on OAC compared with those on SAPT (3.7% vs 2.3%; OR: 1.64; 95% CI: 0.99-2.72; $P = 0.05$) but not compared with those on DAPT (3.7% vs 3.4%; OR: 1.10; 95% CI: 0.66-1.81; $P = 0.72$).

CLINICAL OUTCOMES IN PATIENTS WITH BLEEDING. Table 2 presents clinical outcomes in patients with major and minor bleeding. Patients with major bleeding had a higher mortality risk during the first 30 days (14.1% vs 4.3%; OR: 3.66; 95% CI: 3.11-4.31; $P < 0.001$) and during 1-year follow-up (27.8% vs 14.5%; HR: 1.50; 95% CI: 1.41-1.59; $P < 0.001$) compared with those without bleeding. Patients with major bleeding surviving the first 30 days after TAVR remained at increased mortality risk during the first year compared with those without bleeding (HR: 1.20; 95% CI: 1.08-1.32; $P < 0.001$) (Figure 4). However, mortality risk in patients with 30-day major bleeding was no longer increased after they survived the first 3 months (HR: 1.08; 95% CI: 0.94-1.23; $P = 0.28$). Patients with major bleeding had higher rates of other adverse clinical outcomes, including major vascular complications (10.9% vs 0.3%; OR: 42.51; 95% CI: 29.08-62.15; $P < 0.001$), minor vascular complications (37.9% vs 6.2%; OR: 9.20; 95% CI: 7.90-10.73; $P < 0.001$), and a longer hospital stay (9 days [Q1-Q3: 6-15 days] vs 6 days [Q1-Q3: 4-9 days]; $P < 0.001$). In contrast to patients with major bleeding, patients with minor bleeding did not have higher 30-day mortality than those without bleeding (4.0% vs 4.3%; OR: 0.93; 95% CI: 0.67-1.28; $P = 0.61$) or increased 1-year mortality compared with patients without bleeding (16.7% vs 14.5%; HR: 1.11; 95% CI: 0.93-1.32; $P = 0.27$).

ACCESS-SITE BLEEDING. Access-site bleeding represented 70% of major bleeding complications. Non-access-site bleeding was located mostly in the pericardium (45%) but also in the gastrointestinal tract (17%), urogenital tract (10%), or cerebrum (2%). Figure 5 presents the different locations of major bleeding. Figure 6 displays time-to-mortality curves for access-site and non-access-site bleeding. Mortality risk at 30 days was higher in patients with access-site bleeding (13.6% vs 4.3%; OR: 3.49; 95% CI: 2.27-5.35; $P < 0.001$) and non-access-site bleeding (20.8% vs 4.3%; OR: 5.84; 95% CI: 3.88-8.78; $P < 0.001$) compared with those without bleeding complications. There was a trend toward higher

TABLE 2 Clinical Outcomes in Patients With Major, Minor, or No Bleeding at 30 Days After Transfemoral Transcatheter Aortic Valve Replacement

	No Bleeding (n = 20,874)	Major Bleeding (n = 1,545)	OR for Major vs No Bleeding (95% CI)	Minor Bleeding (n = 1,143)	OR for Minor vs No Bleeding (95% CI)
30 d					
Mortality	789 (4.3)	208 (14.1)	3.66 (3.11-4.31)	41 (4.0)	0.93 (0.67-1.28)
Cardiovascular mortality	689 (3.8)	183 (12.4)	3.70 (3.11-4.40)	34 (3.3)	0.89 (0.62-1.26)
Noncardiovascular mortality	100 (0.5)	25 (1.7)	3.44 (2.21-5.36)	7 (0.7)	1.24 (0.58-2.68)
Stroke	440 (2.1)	50 (3.2)	1.55 (1.15-2.09)	35 (3.1)	1.47 (1.03-2.08)
Disabling stroke	123 (0.6)	25 (1.6)	2.80 (1.82-4.32)	10 (0.9)	1.50 (0.79-2.87)
Myocardial infarction	165 (1.0)	26 (2.2)	2.24 (1.48-3.40)	13 (1.5)	1.51 (0.85-2.27)
Permanent pacemaker	2,969 (14.6)	239 (16.8)	1.18 (1.02-1.37)	243 (21.4)	1.59 (1.37-1.84)
New-onset atrial fibrillation	590 (5.6)	87 (9.7)	1.78 (1.41-2.26)	30 (5.6)	1.00 (0.69-1.46)
Major vascular complication	40 (0.3)	95 (10.9)	42.51 (29.08-62.15)	3 (0.4)	1.40 (0.43-4.54)
Minor vascular complication	876 (6.2)	329 (37.9)	9.20 (7.90-10.73)	319 (40.1)	10.12 (8.65-11.85)
Hospital stay, d	6 (4-9)	9 (6-15)	<0.001[b]	8 (6-13)	<0.001[b]
1 y					
Mortality[a]	1,764 (14.5)	300 (27.8)	1.50 (1.41-1.59)	136 (16.7)	1.11 (0.93-1.32)
Stroke	610 (6.5)	59 (7.1)	1.09 (0.83-1.44)	50 (9.4)	1.48 (1.09-2.00)

Values are n (%) unless otherwise indicated. The 30-day follow-up was complete in 20,793 patients (88%). The 1-year follow-up was complete in 14,097 of patients (60%).
[a]HR is reported for 1-year mortality. [b]*P* value for Mann-Whitney *U* test.

mortality risk among patients with non-access-site bleeding compared with those with access-site bleeding at 30 days (20.8% vs 13.6%; OR: 1.67; 95% CI: 0.94-3.00; $P = 0.08$) and at 1 year (36.5% vs 24.8%; OR: 1.74; 95% CI: 0.98-3.08; $P = 0.06$).

PREDICTORS OF BLEEDING. Multivariate predictors of major bleeding within 30 days after TAVR were female sex (OR: 1.61; 95% CI: 1.34-1.93; $P < 0.001$), body mass index (OR: 0.96 per kg/m^2; 95% CI: 0.95-0.98; $P < 0.001$); peripheral vascular disease (OR: 1.52; 95% CI: 1.20-1.93; $P < 0.001$), and the use of older generation valves (OR: 2.05; 95% CI: 1.63-2.35; $P < 0.001$).

DISCUSSION

The main findings of the present study on bleeding events in >20,000 patients undergoing transfemoral TAVR between 2007 and 2022 are the following: 1) major and minor bleeding occurred in 6.6% and 4.7% of patients undergoing transfemoral TAVR; 2) the incidence of major bleeding declined over time from 2007 to 2022; 3) the use of DAPT, but not OAC, was associated with higher rates of major bleeding; 4) major bleeding was associated with increased mortality risk at both short- and long-term follow-up, whereas minor bleeding did not affect survival after TAVR; and 5) female sex and peripheral vascular disease are important risk factors for major bleeding in patients undergoing transfemoral TAVR. The **Central Illustration** summarizes the main findings.

INCIDENCE OF BLEEDING. The first pivotal clinical trials comparing TAVR with conventional surgery reported significantly lower rates of major bleeding in patients undergoing TAVR. Despite this advantage over conventional surgery, major bleeding was promptly recognized as one of the most important periprocedural complications in patients undergoing TAVR.[8] Our study in a contemporary worldwide cohort confirmed that major bleeding is still frequent and associated with poor outcomes in patients undergoing transfemoral TAVR. National registries from Europe and Asia reported major bleeding rates as high as 19% in patients undergoing TAVR but included primarily earlier generation valves.[9,10] Importantly, we found that the incidence of major bleeding significantly declined worldwide from 2007 to 2022. These results are in line with the recent data from the Society of Thoracic Surgeons/American College of Cardiology TVT (Transcatheter Valve Therapy) Registry on patients undergoing TAVR in the United States between 2011 and 2019.[2] The observed decline in bleeding events shown by our study may be ascribed to various factors, including increased operator's experience and multiple procedural advancements. The latter is confirmed by our finding that older generation valves were associated with higher rates of major bleeding complications.

We observed lower rates of both major and minor bleeding in patients with low surgical risk compared with those with high and intermediate risk profiles. These findings confirm the results of several

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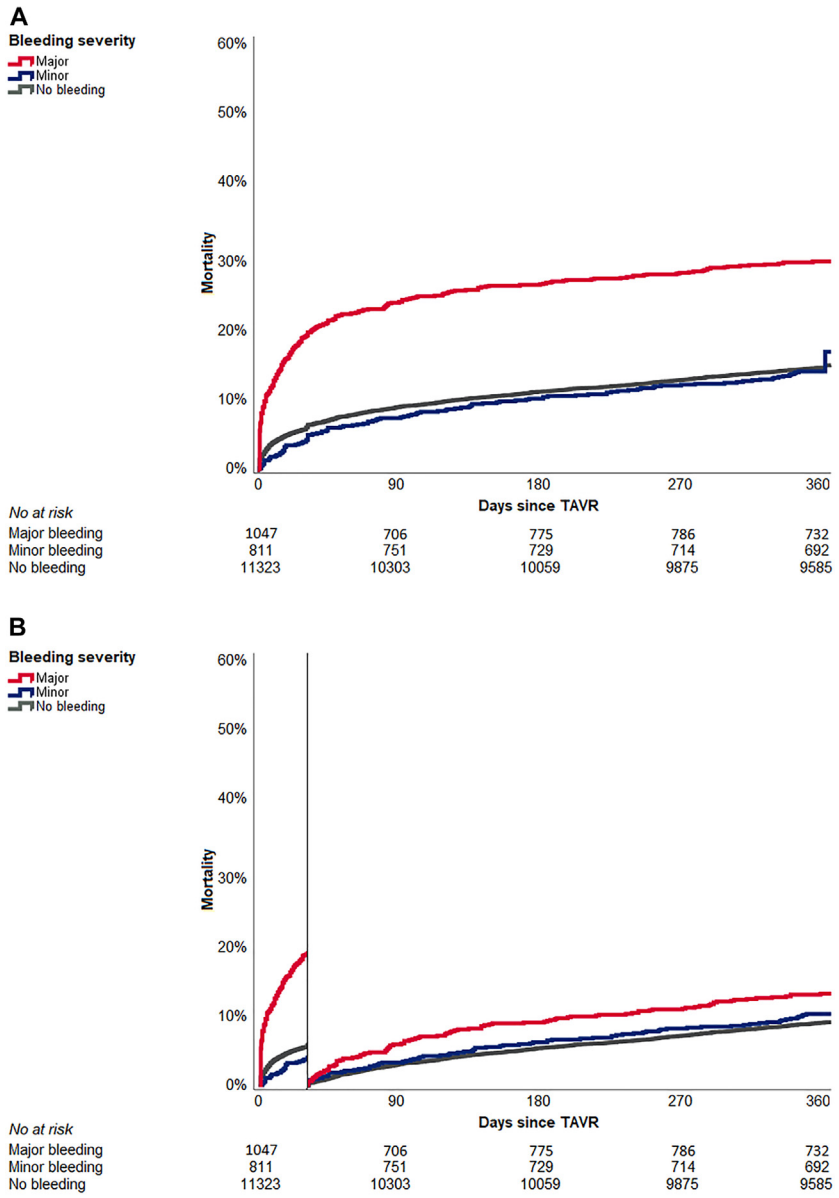
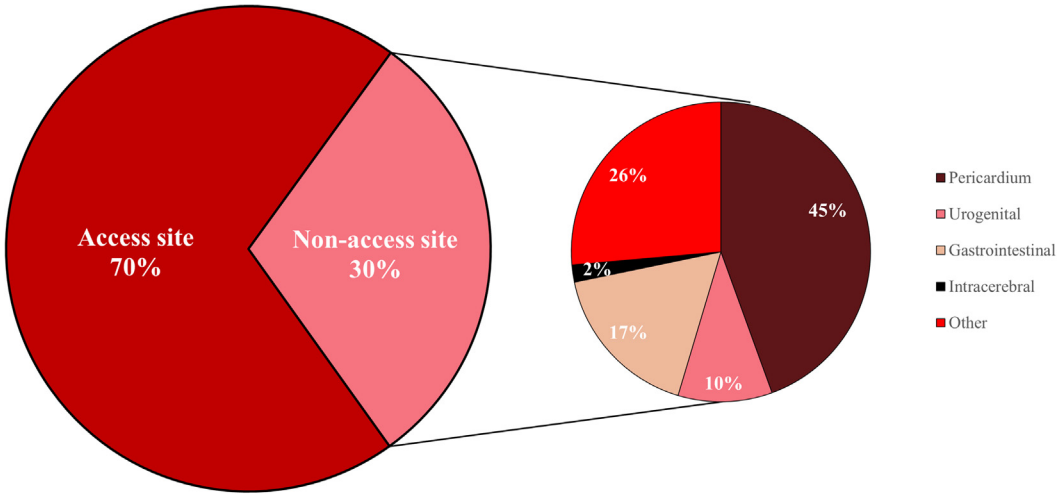
FIGURE 4 Time-to-Mortality Curves of Patients With Major Bleeding or Minor Bleeding and Without Bleeding After Transfemoral TAVR



Time to mortality curves (A) without and (B) with landmark at 30 days for patients with major bleeding, minor bleeding, and without bleeding after transfemoral transcatheter aortic valve replacement (TAVR).

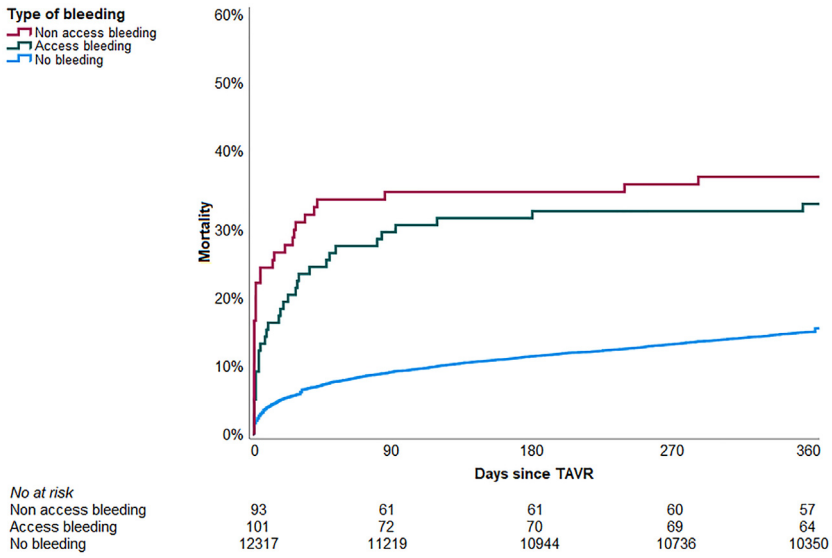
randomized controlled trials in patients with low surgical risk profiles.[3,11,12] However, our results are of particular interest, as randomized controlled trials may not be representative of all contemporary TAVR patients because of selection bias. Furthermore, our results emphasize that the use of DAPT increases the risk for major bleeding complications. This finding may be of great importance for patients with severe aortic valve stenosis and concomitant significant coronary artery disease, as preprocedural PCI necessitates the use of DAPT during TAVR. The optimal timing of coronary revascularization in patients planned to undergo TAVR is being investigated in multiple ongoing studies, including PRO-TAVI (Percutaneous

FIGURE 5 Distribution of the Location of 30-Day Major Bleeding Events in Patients Undergoing Transfemoral Transcatheter Aortic Valve Replacement



Coronary Intervention Before Transcatheter Aortic Valve Implantation; NCT05078619), NOTION-3 (Revascularization in Patients Undergoing Transcatheter Aortic Valve Implantation; NCT03058627), and PCI-TAVI (Optimal Timing of Transcatheter Aortic Valve Implantation and Percutaneous Coronary Intervention—The TAVI PCI Trial; NCT04310046). Interestingly, we did not find increased risk for major bleeding among patients on OAC therapy, which may be due to the periprocedural interruption of antithrombotic therapy at all centers. Several studies, including POPular PAUSE TAVI (Periprocedural Continuation Versus Interruption of Oral Anticoagulant Drugs During Transcatheter Aortic Valve

FIGURE 6 Time-to-Mortality Curves for Patients With Major Access-Site Bleeding, Major Non-Access-Site Bleeding, and No Bleeding After Transfemoral TAVR



TAVR = transcatheter aortic valve replacement.

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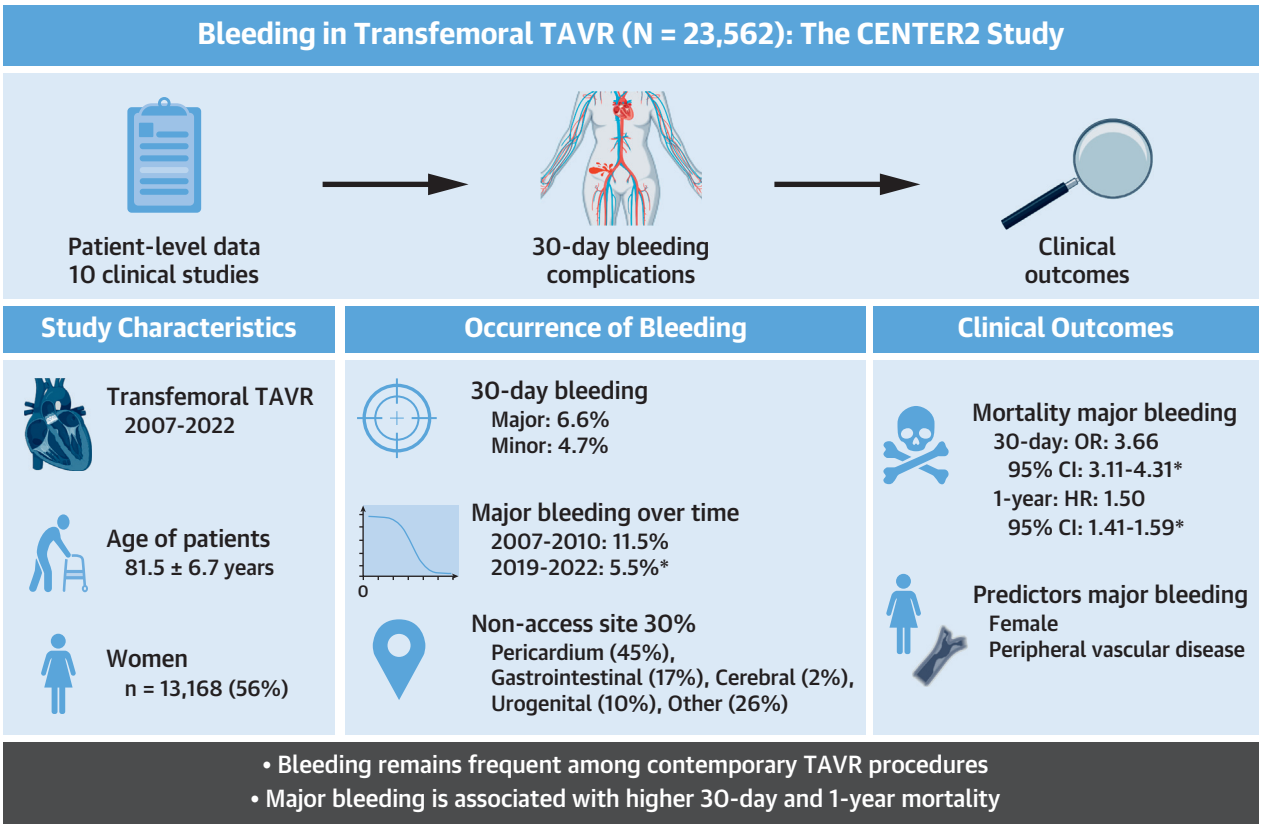
CENTRAL ILLUSTRATION **Bleeding Associated With Transfemoral TAVR**



van Nieuwkerk AC, et al. J Am Coll Cardiol Intv. 2023;16(24):2951–2962.

*$P_{trend} < 0.05$. TAVR = transcatheter aortic valve replacement.

Implantation; NCT04437303), will provide answers on the optimal periprocedural antithrombotic regimen considering both the thrombotic and bleeding risk of patients undergoing TAVR.

CLINICAL OUTCOMES IN PATIENTS WITH MAJOR BLEEDING. To our knowledge, the impact of major bleeding on clinical outcomes has never been evaluated in a large contemporary cohort with more than 20,000 patients undergoing transfemoral TAVR. Our study demonstrated a significantly higher mortality risk for patients with major bleeding during short- and long-term follow-up. Interestingly, major bleeding still had a detrimental impact on 1-year mortality in patients surviving the initial post-procedural period. These findings are in line with the results from randomized controlled trials. Results from patients who were included in the PARTNER (Placement of Transcatheter Aortic Valve) 1 and 2 trials showed that those with major bleeding had a 1.5-fold higher risk for mortality during 1-year follow-up.[13] In contrast to major bleeding, our study did not show a prognostic adverse effect of minor bleeding in patients undergoing transfemoral TAVR. Additionally, our study revealed a significant increase in length of stay in patients who had either a major or minor bleeding after transfemoral TAVR. Prolonged hospitalization after TAVR contributes significantly to increased expenses in patients with bleeding. Results from European registries showed a similar association between periprocedural complications, including major bleeding, and late discharge after TAVR.[14-16] Importantly, results from the PARTNER trial showed that major bleeding is a costly event, with an incremental cost of more than $32,000.[17]

Additionally, both access-site and non-access-site bleeding complications had adverse prognostic

effects on survival after TAVR. However, major bleeding complications that did not originate from the access site of the TAVR procedure tended to have a greater impact on mortality risk compared with access-site bleeding. A Swiss study actually revealed a significant association between the location of the bleeding and mortality.[5] The investigators observed a 1.5-fold higher risk for mortality in patients with non-access-site bleeding compared with those with access-site bleeding. In our study, a substantial proportion of non-access-site bleeding complications were located in the pericardium and the brain, both associated with impaired survival after TAVR.[18]

PREDICTORS OF MAJOR BLEEDING. Our study identified female sex and peripheral vascular disease as predictors of major bleeding in contemporary TAVR patients. In comparison with men, women more often have small and tortuous iliofemoral arteries. Various studies have shown that larger sheath-to-iliofemoral artery ratios are associated with increased rates of vascular complications, frequently leading to access-site bleeding.[19,20] Therefore, the female vascular anatomy may play an important role in the occurrence of major bleeding. In addition, patients with peripheral vascular disease have more severe arterial calcifications and tortuosity. Preprocedural assessment of the iliofemoral tract by computed tomographic angiography is of even greater importance in this subset of patients.

FUTURE STUDIES AND IMPLICATIONS. Our study stresses the importance of the prevention of major bleeding in patients undergoing transfemoral TAVR. Preprocedural identification of patients at risk may be helpful for TAVR operators, as they could implement multiple preventive strategies to avoid bleeding complications. Currently, it is unknown whether bleeding can be profoundly attributed to the procedure or the patient. Frailty has been suggested as an important predictor for major bleeding and mortality in older patients undergoing PCI.[21] Interestingly, studies on frailty and its impact on major bleeding in TAVR patients remain limited.[22] Therefore, future research should investigate the association between frailty and major bleeding, as well as potential interventions to minimize preprocedural frailty. Moreover, future research should also focus on periprocedural antithrombotic management and lean TAVR procedures, including single-access procedures and radial approaches for the non-TAVR access site. Moreover, patient-tailored heparin and protamine dose needs to be further investigated in randomized controlled trials. In addition to overt bleeding complications, future studies should aim to elucidate the importance of so-called nonovert bleeding. Nonovert bleeding is often underreported, as it is characterized by sudden drops in hemoglobin after TAVR without an obvious bleeding source despite extensive diagnostic testing. Undertreatment of nonovert bleeding may contribute to increased mortality risk after TAVR.

STUDY LIMITATIONS. First, the observational design of our study has well-known inherent limitations. Second, multiple registries were included in our study, which may have led to underreporting of clinical outcomes, especially late bleeding events. Clinical outcomes were site reported, and there was no central event adjudication. Moreover, the requirement for blood transfusions was not available for all patients. Periprocedural blood loss or drop in hemoglobin was not captured routinely.

CONCLUSIONS

Bleeding is a frequent and important periprocedural complication in patients undergoing transfemoral TAVR despite decreasing incidence since 2007. Patients with major bleeding after TAVR harbor increased mortality risk at both short- and long-term follow-up, whereas minor bleeding does not affect survival. Female sex and peripheral vascular disease were identified as predictors of major bleeding in transfemoral TAVR. Future studies should focus on validation of risk stratification models and the implementation of preventive strategies.

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PERSPECTIVES

WHAT IS KNOWN? Transfemoral TAVR is associated with lower rates of bleeding compared with conventional surgical aortic valve replacement, but it is still among the most frequent periprocedural complications.

WHAT IS NEW? Despite important advancements over the past decade, bleeding is frequent and potentially fatal in patients undergoing transfemoral TAVR. Major bleeding has an adverse prognostic impact not only on short-term outcomes but also after this initial postprocedural period. Minor bleeding does not affect mortality. Women and patients with peripheral vascular disease harbor the greatest risk.

WHAT IS NEXT? Future studies should focus on preventive strategies to decrease the incidence of major bleeding in patients undergoing transfemoral TAVR, including both preprocedural and periprocedural interventions.

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KEY WORDS aortic valve stenosis, bleeding, mortality, transcatheter aortic valve replacement

APPENDIX For supplemental tables, please see the online version of this paper.